

Sol pc5 f3

1.-

- (a) The field produced by the solenoid in its interior is given by

$$\vec{B} = \mu_0 n I (-\hat{i}) = (4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}) \left(\frac{30.0}{10^{-2} \text{ m}} \right) (15.0 \text{ A}) (-\hat{i})$$

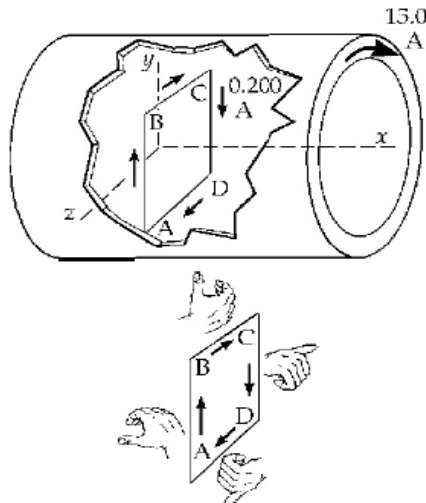
$$\vec{B} = -(5.65 \times 10^{-2} \text{ T}) \hat{i}$$

The force exerted on side AB of the square current loop is

$$(\vec{F}_B)_{AB} = I \vec{L} \times \vec{B} = (0.200 \text{ A})$$

$$\times \left[(2.00 \times 10^{-2} \text{ m}) \hat{j} \times (5.65 \times 10^{-2} \text{ T}) (-\hat{i}) \right]$$

$$(\vec{F}_B)_{AB} = (2.26 \times 10^{-4} \text{ N}) \hat{k}$$



ANS. FIG. P30.43

Similarly, each side of the square loop experiences a force, lying in the plane of the loop, of

$$226 \mu\text{N directed away from the center of the loop}.$$

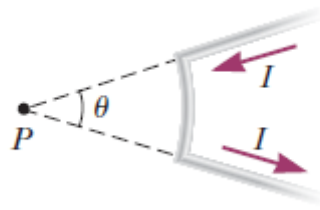
- (b) From the above result, it is seen that the net torque exerted on the square loop by the field of the solenoid is zero. More formally, the magnetic dipole moment of the square loop is given by

$$\vec{\mu} = I \vec{A} = (0.200 \text{ A}) (2.00 \times 10^{-2} \text{ m})^2 (-\hat{i}) = -80.0 \mu\text{A} \cdot \text{m}^2 \hat{i}$$

The torque exerted on the loop is then

$$\vec{\tau} = \vec{\mu} \times \vec{B} = (-80.0 \mu\text{A} \cdot \text{m}^2 \hat{i}) \times (-5.65 \times 10^{-2} \text{ T} \hat{i}) = \boxed{0}$$

2.-Una trayectoria de corriente con la forma que se muestra en la figura 1 produce un campo magnético en P, el centro del arco. Si el arco subtende un ángulo de $\theta = 30.0^\circ$ y el radio del arco es 0.600 m, ¿cuáles son la magnitud y la dirección del campo producido



en P si la corriente es de 3.00 A?

Fig 1

Sol

We use the Biot-Savart law. For bits of wire along the straight-line sections, $d\vec{s}$ is at 0° or 180° to $\hat{r}$, so $d\vec{s} \times \hat{r} = 0$. Thus, only the curved section of wire contributes to $\vec{B}$ at P. Hence, $d\vec{s}$ is tangent to the arc and $\hat{r}$ is radially inward; so $d\vec{s} \times \hat{r} = |ds| 1 \sin 90^\circ \otimes = |ds| \otimes$. All points along the curve are the same distance $r = 0.600$ m from the field point, so

$$B = \int_{\text{all current}} |d\vec{B}| = \int \frac{\mu_0 I}{4\pi} \frac{|d\vec{s} \times \hat{r}|}{r^2} = \frac{\mu_0 I}{4\pi r^2} \int |ds| = \frac{\mu_0 I}{4\pi r^2} s$$

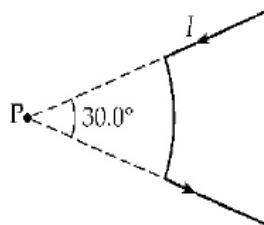
where s is the arc length of the curved wire,

$$s = r\theta = (0.600 \text{ m})(30.0^\circ) \left(\frac{2\pi}{360^\circ} \right) = 0.314 \text{ m}$$

Then,

$$B = (10^{-7} \text{ T} \cdot \text{m/A}) \frac{(3.00 \text{ A})}{(0.600 \text{ m})^2} (0.314 \text{ m})$$

$$B = \boxed{262 \text{ nT into the page}}$$



3.-

- (a) From Ampère's law, the magnetic field at point a is given by $B_a = \frac{\mu_0 I_a}{2\pi r_a}$, where I_a is the net current through the area of the circle of radius r_a . In this case, $I_a = 1.00$ A out of the page (the current in the inner conductor), so

$$B_a = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1.00 \text{ A})}{2\pi(1.00 \times 10^{-3} \text{ m})}$$

$$= \boxed{200 \text{ } \mu\text{T toward top of page}}$$

- (b) Similarly at point b : $B_b = \frac{\mu_0 I_b}{2\pi r_b}$, where I_b is the net current through the area of the circle having radius r_b . Taking out of the page as positive, $I_b = 1.00 \text{ A} - 3.00 \text{ A} = -2.00 \text{ A}$, or $I_b = 2.00 \text{ A}$ into the page. Therefore,

$$B_b = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(2.00 \text{ A})}{2\pi(3.00 \times 10^{-3} \text{ m})}$$

$$= \boxed{133 \text{ } \mu\text{T toward bottom of page}}$$

4.-

IDENTIFY: Calculate the flux through the loop and apply Faraday's law.

SET UP: To find the total flux integrate $d\Phi_B$ over the width of the loop. The magnetic field of a long straight wire, at distance r from the wire, is $B = \frac{\mu_0 I}{2\pi r}$. The direction of $\vec{B}$ is given by the right-hand rule.

EXECUTE: (a) $B = \frac{\mu_0 i}{2\pi r}$, into the page.

b)

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = \frac{\mu_0 L}{2\pi} \ln(b/a) \frac{di}{dt}.$$

c)

$$|\mathcal{E}| = \frac{\mu_0 (0.240 \text{ m})}{2\pi} \ln(0.360/0.120)(9.60 \text{ A/s}) = 5.06 \times 10^{-7} \text{ V}.$$